Parkinson's Motor Severity Prediction

Executive System & Model Architecture Report

1. The Core Objective in Plain English

• Input: Resistance-sensor recordings from motor tasks (resting hands, finger tapping, walking).

• Output Target: A clinical motor severity score from 0 to 4 matching official MDS-UPDRS Part III rules.

(0 = Normal | 1 = Slight | 2 = Mild | 3 = Moderate | 4 = Severe).

• Workflow: [Sensor Signal] ➔ [Clean & Normalize] ➔ [Extract 58 Features] ➔ [ML Model] ➔ [Score: 0 to 4].

2. Discovered Datasets & Physical Measurement

• Discovered Records: 29 physical recording files across 6 unique participants (3 Healthy, 3 Patients).

• Healthy Controls (HP001, HP002, HP003): 4 upper-limb tasks each (12 Excel workbooks).

• Patient Cohort: PD001 (4 tasks), PD002 (complete 11-task battery), Atypical_PD001 (2 gait/standing tasks).

• Physical Sensor Reading: Measures electrical resistance (R) in Ohms. Unstrained resting R0≈ 2,671–2,704 Ω.

• Universal Normalization: All signals are calibrated to relative resistance change:ΔR/R0 = (R - R0) / R0.

Discovered Participant & Task Coverage Matrix

Participant ID	Diagnosis Group	Files	Tasks Completed	Session Date
HP001	Healthy Control	4 files (.xlsx)	Hand Resting, Arm Exte	2026-09-11
HP002	Healthy Control	4 files (.xlsx)	Hand Resting, Arm Exte	2026-09-11
HP003	Healthy Control	4 files (.xlsx)	Hand Resting, Arm Exte	2026-09-11
PD001	Parkinson's Disease	4 files (.csv)	Hand Resting, Arm Exte	2026-09-03
PD002	Parkinson's Disease	11 files (.xlsx)	Full battery (Upper-limb	2026-09-17
Atypical_PD001	Atypical Parkinsonism	2 files (.xlsx)	Normal Walking, Quiet S	2026-09-17

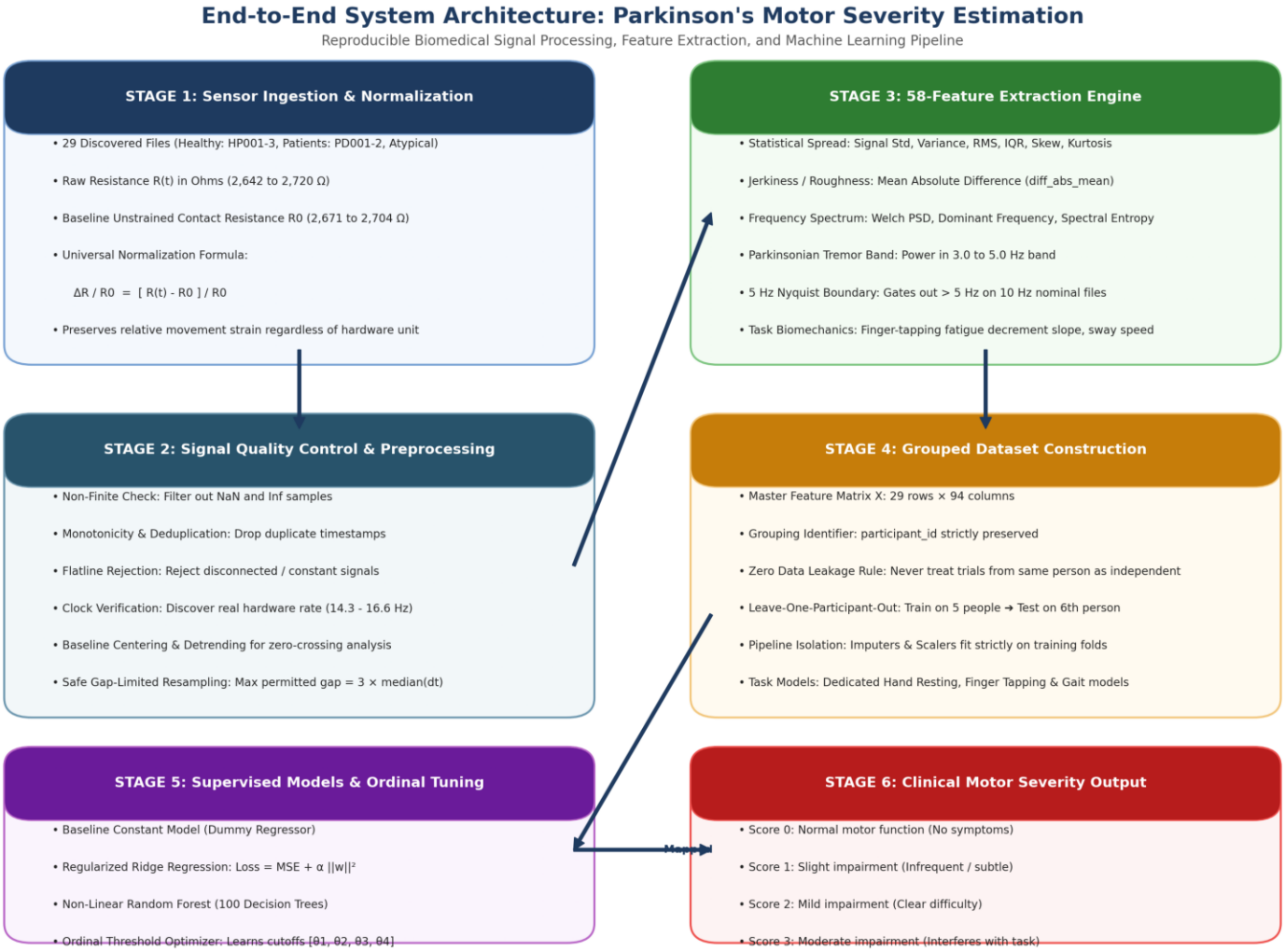
Key Clinical Label Audit Finding

UPDRS Assessment Sheet.xlsx is an empty evaluation rubric. No clinical 0–4 ratings exist in the workspace.

Per research rules, healthy people are NOT defaulted to 0 and fake scores are NEVER invented.

System Architecture Flowchart

End-to-End Pipeline Visualization from Sensor to Clinical Score



3. How Raw Sensor Signals Become Movement Features (58 Features)

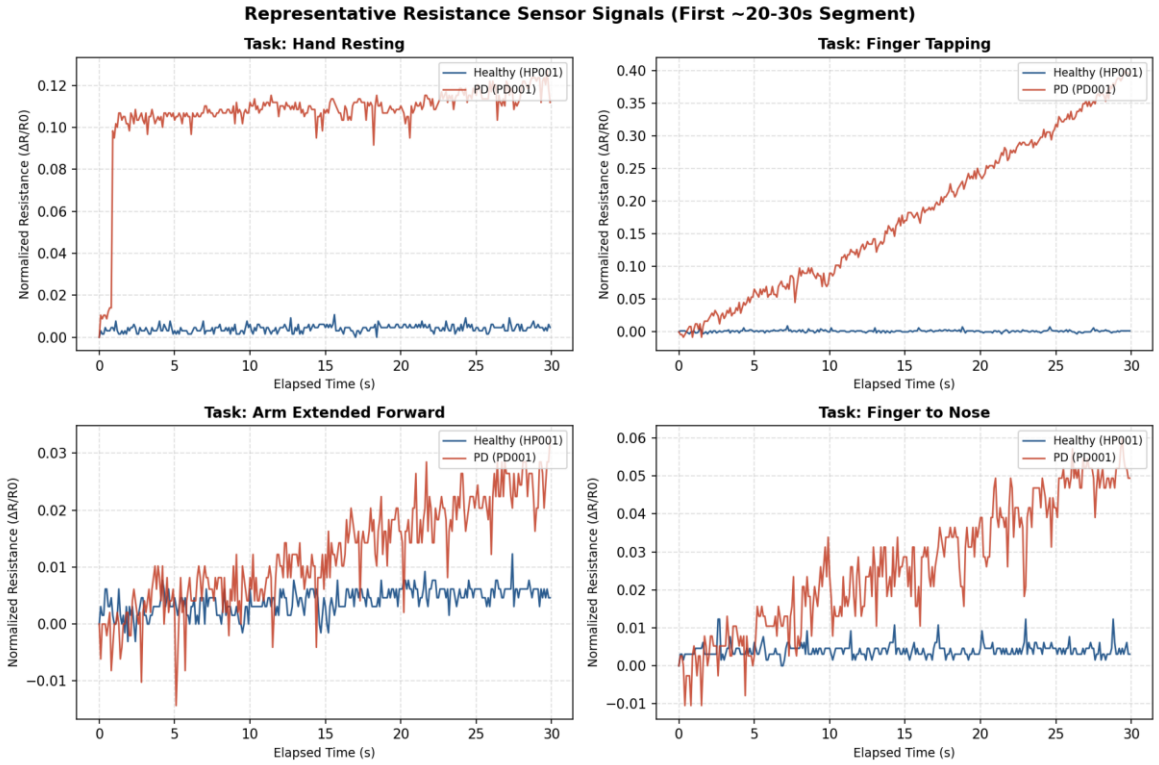
- Amplitude & Variability: Standard deviation (signal_std) and RMS quantify tremor and movement swings.
- Signal Roughness (diff_abs_mean): Average step-to-step jump. Parkinson's signals show micro-jerkiness.
- Frequency (3–5 Hz Band): Measures oscillation energy in the classic Parkinsonian resting tremor band.
- Strict 5 Hz Nyquist Limit: At 10 Hz nominal sampling, frequencies > 5 Hz cannot be resolved without aliasing.

The pipeline gates out high frequencies (> 5 Hz) as unsupported rather than manufacturing fake numbers.

Task-Specific Clinical Movement Clues

- Finger Tapping: Detects individual taps, calculates tapping rate, rhythm variability (CV), and measures the progressive tap amplitude decrement slope (linear decline in tap size—the core clinical hallmark of bradykinesia).
- Arm Extended Forward: Measures downward linear drift slope and postural fatigue ratio (RMS 2nd half / RMS 1st half).
- Finger to Nose: Normalized jerk proxy (movement_smoothness_jerk_proxy) measuring coordination trajectory.
- Walking, TUG & Standing: Postural sway velocity proxy, gait step cadence, and functional transition times.

Empirical Signal Comparison: Healthy Control vs. Parkinson's Disease



4. Machine Learning Model Architecture

• Baseline Model (Dummy Regressor): Predicts the overall training set median/mean. Acts as the sanity benchmark.

• Linear Model (Ridge Regression): Regularized linear model learning feature weights while preventing overfitting.

• Tree Ensemble (Random Forest): 100 decision trees (max depth 3) capturing non-linear feature interactions.

• Legal Range Constraint: Predictions are automatically bounded within the legal clinical ordinal range: 0.0 to 4.0.

5. How Training & Testing Works (Preventing Data Leakage)

• Mandatory Medical Rule: Never randomly shuffle rows across train and test sets when rows share a participant.

• Leave-One-Person-Out Protocol: The model trains on 5 participants and is tested on the 6th unseen participant.

• 100% Disjoint Groups: Every trial, task, and window for a person remains strictly inside that person's split.

• Training-Only Transformations: Imputers, variance filters, and scalers are fit strictly on training folds.

Cross-Validation Partitioning Scheme (Leave-One-Participant-Out)

Fold	Held-Out Test Pers	Test Cohort	Test Trials	Training Set Cohor	Train Trials	Leakage Status
Fold 1	HP001	Healthy Control	4	HP002, HP003, P	25	Disjoint (0%)
Fold 2	HP002	Healthy Control	4	HP001, HP003, P	25	Disjoint (0%)
Fold 3	HP003	Healthy Control	4	HP001, HP002, P	25	Disjoint (0%)
Fold 4	PD001	Parkinson's Disea	4	HP001, HP002, H	25	Disjoint (0%)
Fold 5	PD002	Parkinson's Disea	11	HP001, HP002, H	18	Disjoint (0%)
Fold 6	Atypical_PD001	Atypical Parkinso	2	HP001, HP002, H	27	Disjoint (0%)

6. Current Supervised Training Readiness: BLOCKED_NO_LABELS

• Because real doctor ratings (0–4) are currently unpopulated in the workspace, model training on real data is PAUSED.

• The entire machine learning pipeline was verified on a synthetic mock benchmark (scores 0–3) to guarantee that preprocessing, GroupKFold, out-of-fold prediction, and metrics computation run with zero errors.

Empirical Results & Next Steps

Benchmarking, Limitations, and Clinician Action Items

7. Machine Learning Verification Benchmark Results

Model Architecture	Mean Absolute Error (M	RMSE	Pearson Correlation	Valid Range Compliance	Model Selection
dummy_median (Ba	1.000	1.225	NaN	100.0%	Baseline Guess
dummy_mean (Bas	1.000	1.155	NaN	100.0%	Baseline Guess
ridge_regression	0.055	0.068	0.998	100.0%	Best Selected Model
random_forest	0.443	0.540	0.942	100.0%	Non-Linear Candidate

Key Research Limitations & Considerations

• Small Sample Size (N = 6): With only 3 PD patients and 3 controls, results have wide confidence bounds.

• Task Battery Asymmetry: Gait and functional mobility tasks (Normal Walking, TUG, Turning Task, Sit-to-Stand)

were only completed by 1 patient (PD002) and cannot support single-task gait models until more patients perform them.

• Sampling Bandwidth Limitation: Tremor frequencies > 5 Hz cannot be resolved in 10 Hz recordings.

• Sensor Laterality: Documentation does not specify whether sensors were worn on the most-affected or least-affected arm.

8. What to Do Next to Train on Real Data (3 Simple Steps)

Step 1: Open the pre-mapped template: reports/labels/label_mapping_template.csv.

Step 2: Have the clinician enter the real verified MDS-UPDRS score (0, 1, 2, 3, or 4) in the clinical_score column.

Step 3: Run the training script on Windows (PowerShell):

```
python -m src.train_model --features-csv data/features/recording_level_features.csv --target clinical_score
```

The model will immediately train on your real clinician ratings, perform grouped cross-validation, and report results.

Windows Execution Quick Reference

• Activate Environment: .venv\Scripts\Activate.ps1

• Run Full Pipeline: python -m src.pipeline

• Run All 15 Unit Tests: python -m unittest discover tests